TryHackMe Writeup: Linux Fundamentals Part 1

- **Platform:** TryHackMe
- **Room:** https://tryhackme.com/room/linuxfundamentalspart1
- **Difficulty:** Beginner / Informational
- **Topics:** Linux terminal, navigation, file searching, log analysis, shell operators

1. Objective

This room introduces essential Linux command-line skills. It covers identifying the current user, displaying text, navigating folders, reading files, searching logs, and combining commands or redirecting output.

2. Task and Method

Task 1 - Introduction

Start the TryHackMe lab machine and open the in-browser terminal. The virtual machine is intended for this exercise. Stop or terminate the lab machine after finishing.

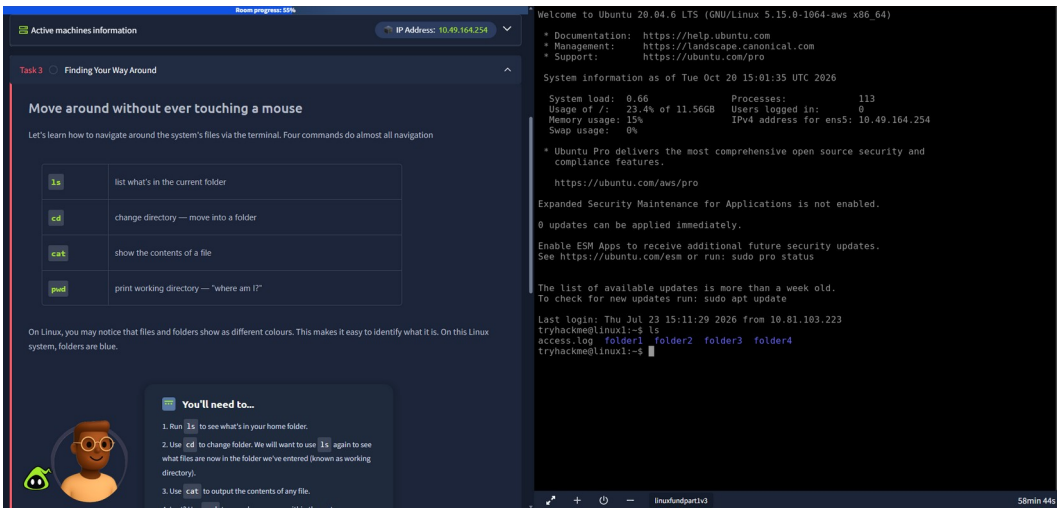
Task 2 - Talking to Linux

Commands

Command	Purpose	Example
whoami	Prints the username of the current effective user.	whoami
echo	Prints text to the terminal.	echo TryHackMe

Steps

1. Run whoami and record the username shown.
2. Run echo TryHackMe.
3. Confirm that the terminal prints TryHackMe.



Answers

- **Command to find the current username:** whoami
- **Command to output text:** echo
- **Username shown by the standard room machine:** tryhackme - verify in your own terminal.

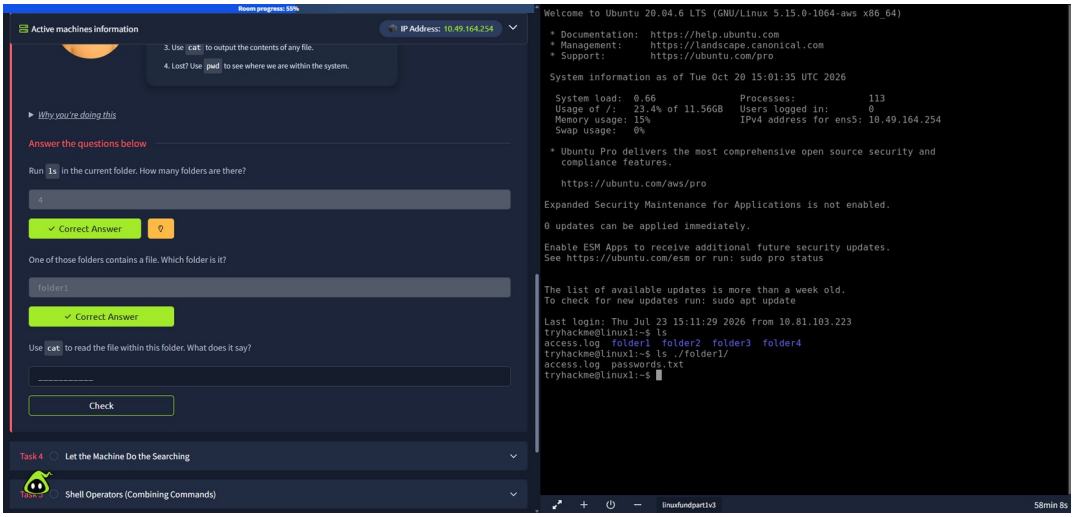
Task 3 - Finding Your Way Around

Commands

Command	Purpose	Example
ls	Lists files and folders.	ls
cd	Changes the current directory.	cd folder4
cat	Displays a file's contents.	cat note.txt
pwd	Prints the current working directory.	pwd

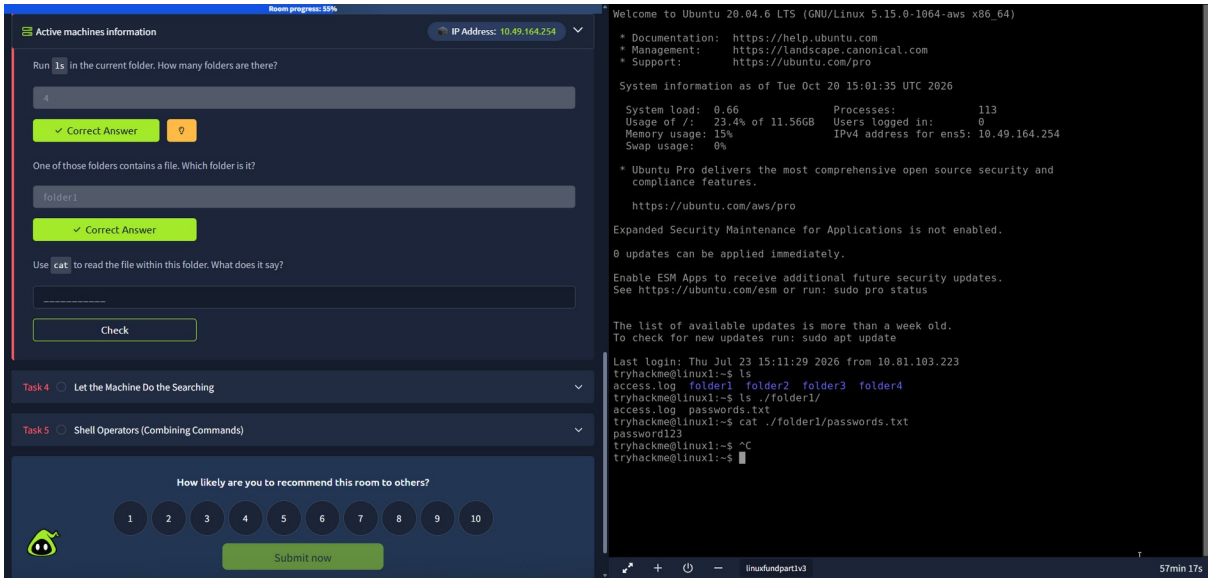
Steps

1. Run ls to list the home directory.
2. Count the folders and identify which folder contains a file.
3. Read the file with cat.
4. Navigate into the folder and run pwd if you want to confirm its full path.



Answers to verify against the live machine

- **Number of folders:** 4
- **Folder containing a file:** folder1
- **File contents:** password123



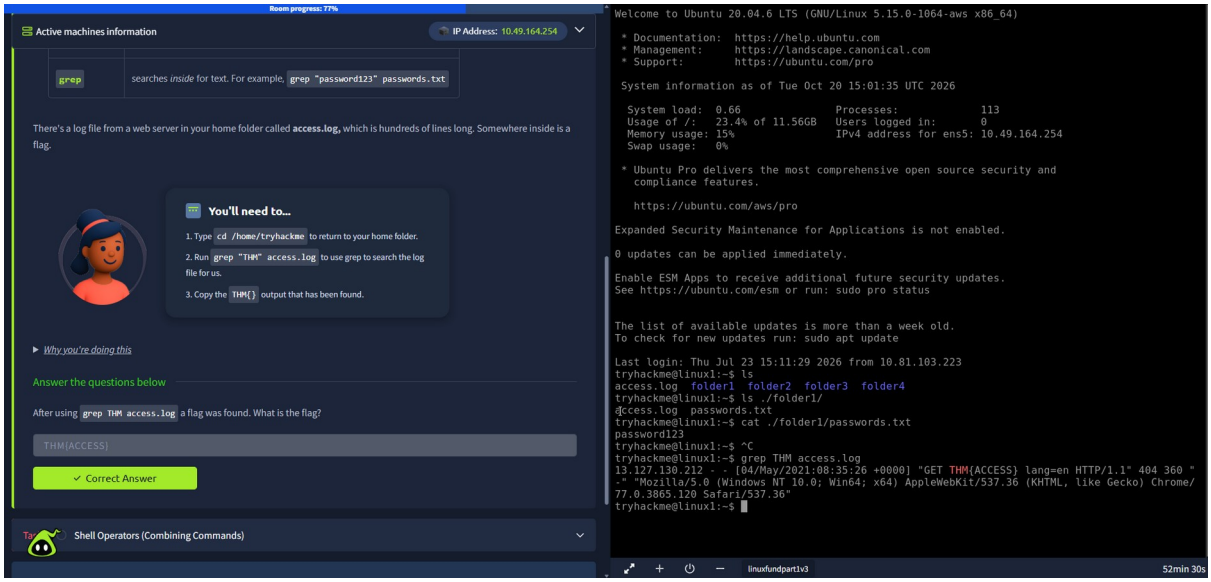
Task 4 - Let the Machine Do the Searching

The grep command searches text for a matching pattern. The room uses a web-server log file called access.log and asks you to find the flag with the prefix THM.

Steps

1. Search the log:

   grep "THM" access.log
2. Record the flag shown by the lab.



Answer

- **Flag reported for this room:** THM{ACCESS}

Security relevance: SOC analysts often search authentication and web-server logs to find indicators, suspicious requests, or unusual activity. Search only logs and systems you are authorized to inspect.

Task 5 - Shell Operators

Shell operators control how commands run and how command output is redirected.

Operator	Meaning	Example
&	Runs a command in the background.	sleep 30 &
&&	Runs the second command only if the first command succeeds.	cd /home/tryhackme && pwd
>	Sends output to a file, replacing existing contents.	echo hello > welcome
>>	Appends output to the end of a file without replacing existing contents.	echo again >> welcome

Practical examples

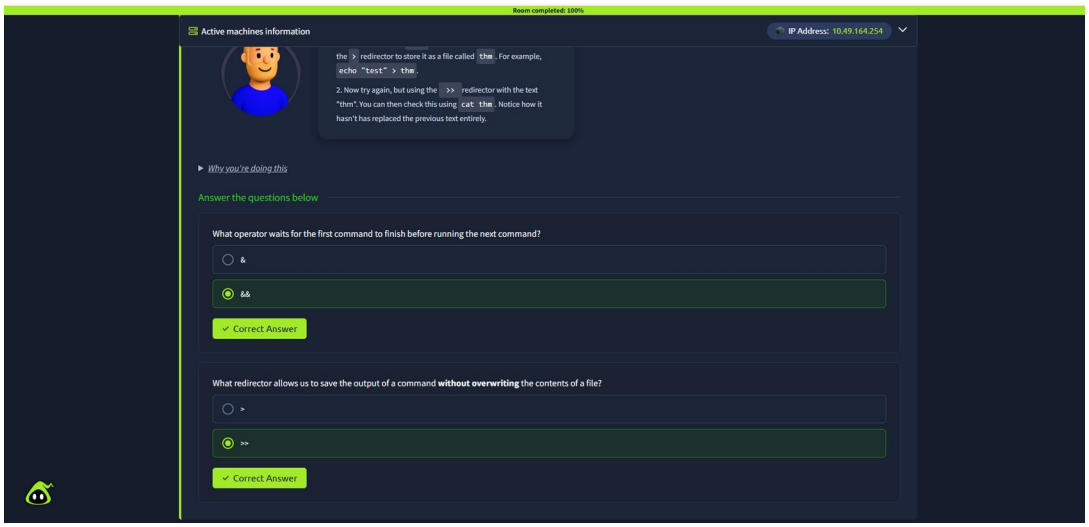
Create or overwrite a file:

```
echo "TryHackMe" > thm
cat thm
```

Append to the same file:

```
echo "thm" >> thm
cat thm
```

After the second command, the file contains both lines.



Answers

- **Operator that waits for the first command to finish before running the next, if the first succeeds:** &&
- **Redirector that appends output without overwriting existing content:** >>

3. Answer Summary

Task	Question	Answer
2	Command to identify the current user	whoami
2	Command to output text	echo
2	Username on the standard room machine	tryhackme - verify
3	Number of folders	4 - verify
3	Folder containing a file	folder4 - verify
3	Contents of the file	Hello World! - verify
3	Path after entering the folder	/home/tryhackme/folder4 - verify if applicable
4	Flag found in access.log	THM{ACCESS} - verify
5	Sequential conditional operator	&&
5	Append-output redirector	>>

4. Key Learning Outcomes

- The terminal provides a text-based way to control a Linux system.
- whoami helps identify the current user; account identity influences access rights.
- ls, cd, cat, and pwd support basic navigation and file inspection.
- grep can quickly locate relevant strings inside large log files.
- && runs a later command only when the preceding command succeeds.
- > overwrites a file, whereas >> appends to it.
- Logs can contain sensitive information, so access and sharing should be handled responsibly.